

RECORD

Field Details

Course	Full Stack Web Development
Course Code	23CM4121
Experiment No.	2
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Date	30-08-2026

1. TITLE

Design a College Website Using HTML5 and CSS3

2. AIM

To design a multi-section college website using HTML5 and CSS3, featuring a styled header, navigation bar, informational sections, a course showcase, and a contact form.

3. OBJECTIVE

The objectives of this experiment are:

- To structure a webpage using HTML5 semantic elements such as header, nav, section and footer.
- To style the page using CSS3 properties such as background colors, padding, border-radius and flexbox.
- To create a responsive-looking course showcase using flexbox (display: flex).
- To apply hover effects on navigation links and course cards.
- To build a contact form using text, email, select and textarea inputs, styled with CSS.
- To style buttons with background colors and hover effects.

4. THEORY

HTML5 provides semantic elements such as <header>, <nav>, <section> and <footer> that describe the role of each part of a page, making the structure easier to read and maintain than a page built only from generic <div> elements.

CSS3 enables visual styling separate from the HTML structure. Properties such as background-color, padding, border-radius and margin control the appearance and spacing of elements, while the Flexbox layout model (display: flex, gap) is used here to arrange the three course cards evenly in a row.

Pseudo-classes such as :hover allow style changes in response to user interaction without JavaScript — used here to highlight navigation links in yellow and to scale up a course card slightly (transform: scale(1.03)) when the mouse hovers over it.

A contact form is built using standard HTML input types (text, email), a <select> dropdown for branch selection, and a <textarea> for the message, all styled uniformly through shared CSS rules for input, select and textarea to give a consistent look.

5. ALGORITHM / PROCEDURE

- Create a new HTML file and set up the basic HTML5 document structure.
- Add an internal <style> block and define base styles for the body, including font family and background color.
- Style the <header> with a background color, centered text and padding, and add the college name and tagline.

- Create a <nav> bar with navigation links; style the links and add a :hover color change.
- Wrap the main content in a .container div to control page width.
- Add a Home <section> with a welcome heading and description.
- Add an About <section> with a description and a bulleted list of college highlights.
- Add a Courses <section> containing a .courses flex container with three .course cards (CSE, ECE, MECH).
- Add a hover effect on .course to slightly scale up the card on mouse-over.
- Add a Contact <section> with a form containing Name, Email, Branch (dropdown) and Message (textarea) fields.
- Style the form's input, select and textarea elements consistently, and style the Submit/Reset buttons with a hover effect.
- Add a <footer> with a centered copyright line.
- Open the HTML file in a browser and verify that all sections, styling and hover effects work correctly.
- Test the contact form by entering sample values in each field.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Website</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background-color: #f4f4f4;
    }
    header {
      background-color: #1e3a5f;
      color: white;
      text-align: center;
      padding: 25px;
    }
    nav {
      background-color: #222;
      padding: 12px;
      text-align: center;
    }
    nav a {
      color: white;
      text-decoration: none;
      margin: 20px;
    }
    nav a:hover {
      color: yellow;
    }
    .container {
      width: 90%;
      margin: auto;
    }
    section {
      background-color: white;
      margin: 20px 0;
      padding: 25px;
      border-radius: 10px;
    }
    h2 {
      color: #1e3a5f;
    }
  </style>
</head>
<body>
  <header>
    <h1>College Website</h1>
  </header>
  <nav>
    <a href="#home">Home</a>
    <a href="#about">About</a>
    <a href="#courses">Courses</a>
    <a href="#contact">Contact</a>
  </nav>
  <div class="container">
    <section id="home">
      <h2>Welcome</h2>
      <p>This is a sample website for a college. It contains information about the college, its courses, and contact details. The website is designed to be responsive and user-friendly.</p>
    </section>
    <section id="about">
      <h2>About Us</h2>
      <p>Our college is a leading institution in the field of engineering and technology. We offer a wide range of undergraduate and postgraduate programs. Our faculty consists of highly qualified and experienced professionals. We are committed to providing quality education and fostering a spirit of innovation and research.</p>
      <ul>
        <li>• We offer a wide range of undergraduate and postgraduate programs.</li>
        <li>• Our faculty consists of highly qualified and experienced professionals.</li>
        <li>• We are committed to providing quality education and fostering a spirit of innovation and research.</li>
      </ul>
    </section>
    <section id="courses">
      <h2>Courses</h2>
      <p>We offer a wide range of undergraduate and postgraduate programs. Our faculty consists of highly qualified and experienced professionals. We are committed to providing quality education and fostering a spirit of innovation and research.</p>
      <div class="courses">
        <div class="course">
          <h3>CSE</h3>
          <p>Computer Science and Engineering</p>
        </div>
        <div class="course">
          <h3>ECE</h3>
          <p>Electronics and Communication Engineering</p>
        </div>
        <div class="course">
          <h3>MECH</h3>
          <p>Mechanical Engineering</p>
        </div>
      </div>
    </section>
    <section id="contact">
      <h2>Contact Us</h2>
      <p>Name: <input type="text"> Email: <input type="text"> Branch: <select></select> Message: <textarea></textarea> Submit <input type="button" value="Submit"/> Reset <input type="button" value="Reset"/>
    </section>
  </div>
  <div class="container">
    <div class="copyright">
      <p>© 2023 College Website. All rights reserved.</p>
    </div>
  </div>
</body>
</html>
```

```

.courses {
display: flex;
gap: 20px;
}
.course {
background-color: #e8f0f8;
padding: 20px;
flex: 1;
text-align: center;
border-radius: 8px;
}
.course:hover {
transform: scale(1.03);
}
form {
max-width: 500px;
margin: auto;
}
input, select, textarea {
width: 100%;
padding: 10px;
margin: 8px 0 15px;
border: 1px solid #ccc;
border-radius: 5px;
}
button {
background-color: #1e3a5f;
color: white;
padding: 10px 20px;
border: none;
border-radius: 5px;
}
button:hover {
background-color: #345f8c;
}
footer {
background-color: #1e3a5f;
color: white;
text-align: center;
padding: 15px;
}
</style>
</head>
<body>
<!-- Header -->
<header>
<h1>ANITS College</h1>
<p>Learning Today, Leading Tomorrow</p>
</header>
<!-- Navigation -->
<nav>
<a href="#home">Home</a>
<a href="#about">About</a>
<a href="#courses">Courses</a>
<a href="#contact">Contact</a>
</nav>
<div class="container">
<!-- Home -->
<section id="home">
<h2>Welcome</h2>
<p>
Welcome to the ANITS College website.
We provide quality education and encourage
students to develop technical and professional skills.
</p>

```

```

</section>
<!-- About -->
<section id="about">
<h2>About Us</h2>
<p>
Our college focuses on academic excellence, innovation,
teamwork and overall student development.
</p>
<ul>
<li>Experienced Faculty</li>
<li>Modern Laboratories</li>
<li>Library Facilities</li>
<li>Sports and Cultural Activities</li>
</ul>
</section>
<!-- Courses -->
<section id="courses">
<h2>Courses Offered</h2>
<div class="courses">
<div class="course">
<h3>CSE</h3>
<p>Computer Science and Engineering</p>
</div>
<div class="course">
<h3>ECE</h3>
<p>Electronics and Communication Engineering</p>
</div>
<div class="course">
<h3>MECH</h3>
<p>Mechanical Engineering</p>
</div>
</div>
</section>
<!-- Contact Form -->
<section id="contact">
<h2>Contact Us</h2>
<form>
<label>Name:</label>
<input type="text" placeholder="Enter your name">
<label>Email:</label>
<input type="email" placeholder="Enter your email">
<label>Branch:</label>
<select>
<option>Select Branch</option>
<option>CSE</option>
<option>ECE</option>
<option>MECH</option>
</select>
<label>Message:</label>
<textarea rows="4" placeholder="Enter your message"></textarea>
<button type="submit">Submit</button>
<button type="reset">Reset</button>
</form>
</section>
</div>
<!-- Footer -->
<footer>
<p>&copy; 2026 ANITS College | All Rights Reserved</p>
</footer>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<header>, <nav>, <section>, <footer>	Semantic HTML5 elements used to structure the banner, navigation, content blocks and footer.
background-color / padding / border-radius	CSS3 properties used to color, space out and round the corners of sections, cards and buttons.
display: flex; gap: 20px;	Arranges the three course cards in a single row with equal spacing using Flexbox.
nav a:hover	Changes navigation link color to yellow when the mouse hovers over it.
.course:hover { transform: scale(1.03); }	Slightly enlarges a course card on hover for a subtle interactive effect.
<select> / <option>	Provides a dropdown for the user to choose their branch (CSE, ECE, MECH) in the contact form.
<textarea rows="4">	Provides a multi-line input for the user's message in the contact form.
button:hover	Changes the Submit/Reset button background color on hover for visual feedback.
input, select, textarea { width: 100%; ... }	Applies consistent width, padding, border and spacing to all form controls.

8. TEST CASES AND EXECUTION DATA

Each section and interactive element was verified by loading the page in a browser.

Test Case	Feature Tested	Expected Result	Status
TC-01	Page load	Header, nav, Home, About, Courses and Contact sections render correctly	Pass
TC-02	Navigation link hover	Link text turns yellow when hovered	Pass
TC-03	Course cards layout	CSE, ECE and MECH cards appear in a single row with equal width	Pass
TC-04	Course card hover	Card scales up slightly when hovered	Pass
TC-05	Contact form fields	Name, Email, Branch dropdown and Message textarea accept input	Pass
TC-06	Submit/Reset button hover	Button background color changes on hover	Pass

9. OUTPUT

Output 1: Home, About and Courses Sections

The page displays the styled header and navigation bar, followed by the Welcome and About Us sections, and the Courses Offered section with three evenly spaced course cards.

The screenshot shows a web browser displaying the ANITS College website. The browser's address bar shows the file path: `file:///C:/Users/vamshy/OneDrive/Desktop/html/exp%202.html`. The website has a dark blue header with the text "ANITS College" and the tagline "Learning Today, Leading Tomorrow". Below the header is a navigation bar with links: Home, About, Courses, and Contact. The main content area is divided into several sections: a "Welcome" section with a brief introduction, an "About Us" section listing college features like experienced faculty and modern laboratories, and a "Courses Offered" section displaying three course cards: CSE (Computer Science and Engineering), ECE (Electronics and Communication Engineering), and MECH (Mechanical Engineering). At the bottom is a "Contact Us" section containing a form with fields for Name, Email, Branch (a dropdown menu), and Message, along with Submit and Reset buttons.

Output 2: Contact Form Filled

The contact form accepts user input, with the Name, Email and Branch dropdown filled in and the Submit/Reset buttons ready to be used.

This screenshot shows the same "Contact Us" form as the previous image, but now it is filled with user input. The "Name" field contains "R Abhiram", the "Email" field contains "rowthubhaskarabhiram24.csm@anits.edu.in", and the "Branch" dropdown menu is set to "CSE". The "Message" field is empty, showing the placeholder text "Enter your message". The Submit and Reset buttons remain at the bottom of the form.

10. RESULT

Thus, a multi-section college website was successfully designed using HTML5 and CSS3, featuring a styled header, navigation bar with hover effects, informational sections, a flexbox-based course showcase with hover interactions, and a styled contact form. All sections and interactive elements were verified in the browser and worked as expected.